

Agentic RAG: Multi-Project Research Lab

Knowledge Base – Technical Implementation Report

1. Executive Summary

This report documents the design and implementation of a secure, multi-tenant Retrieval-Augmented Generation (RAG) system created for a research center hosting four specialized laboratories: U1 Genomics, U2 NLP, U3 Robotics, U4 Materials Science, plus shared public documentation.

The system uses a defense-in-depth security strategy with injection detection, strict access control, and PII masking, while supporting conversational memory via the LangChain framework.

Key Results:

- Security: 100% red-team attack blocking
- Accuracy: 75% overall test pass rate with 75% citation fidelity
- Compliance: 100% PII masking (CNIC/phone patterns fully suppressed)
- Performance: ~2.3 seconds average query latency

2. System Architecture

2.1 Data Flow Summary

1. Input Validation: Planner detects injection or malicious intent
2. Retrieval: Vector search across tenant-scoped + public namespaces
3. Access Control: Policy guard enforces tenant isolation
4. Privacy Protection: Regex-based PII masking
5. Context Enhancement: Conversation memory appended
6. Answer Generation: LLM responds with citations
7. Persistence: Masked turns stored
8. Audit Logging: Every request logged

3. Planner Logic & Injection Detection

3.1 Security-First Design with two detection categories.

3.1.1 Injection Patterns (20 total)

Examples include: "ignore all previous", "override policy_guard", "bypass policy", "dump memory", "exfiltrate", "unmask", "without any guards".

Action: Refusal → InjectionDetected

3.1.2 Prohibited Patterns (18 total)

Examples include: "list phone numbers", "dump cnic", "cross-tenant", "reveal confidential".

Action: Refusal → LeakageRisk

4. Policy Guard & PII Protection

Access Control Matrix ensures tenant isolation with PII masking applied on all allowed queries.

4.2 PII Masking Regex

CNIC: \b\d{5}-\d{7}-\d\b

Phone: \+?92-?\d{3}-?\d{7}

Masking applied at retrieval, memory saving, and output.

5. Memory Design & Token Cost

Comparison between Buffer Memory and Summary Memory with trade-offs in storage, context size, and recall quality.

6. Test Results & Red-Team Evaluation

All unit tests passed.

10/10 red-team attacks blocked.

Evaluation harness: 7/8 passes with one failure due to missing U3-specific document content.

Citation fidelity at 75%.

Performance Metrics:

- Red-team blocking: 100% (Pass)
- Citation fidelity: 75% (Below target)

- PII leakage: 0% (Pass)
- Cross-tenant leakage: 0% (Pass)
- Average latency: 2.3s (Pass)